STRUCTURAL INTEGRITY & LOAD AUDIT: NORTH RIVER VIADUCT SPAN B

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1. Dead & Live Load Superposition Summary

This structural assessment evaluates Span B of the North River Viaduct according to highway bridge design code [1] and seismic retrofit guidelines [2]. Ultrasonic weld measurment confirmed that primary box-girder flanges meet the required structural tolerence.

Unfactored vertical reaction force at Pier P3 was computed by summing dead, wearing surface, and live traffic loads: 1850.0 + 420.0 + 680.0 = 3010.0 kN at mid-span bearing pads.

Peak wind gust velocity across the deck reached 28.5 mps during storm monitoring, while backup de-icing heaters consumed 18.4 MW/h over the winter test period. Geotechnical borehole logs are included in Appendix A, and fatigue crack propagation curves are detailed in Appendix D alongside finite-element verification in citation [5].

2. Reinforcement & Bearing Retrofit Bill of Materials

Component Description	Quantity (Units)	Unit Rate ($)	Extended Cost ($)
Elastomeric Bearing Pad	16	450.0	7200.0
High-Strength Anchor Bolt	40	85.0	3600.0
Carbon Fiber Wrap Kit	12	600.0	7200.0
Epoxy Grout Injection	25	120.0	3000.0
Total	**93**	**1255.0**	**21450.0**

3. Post-Tensioning Cable & Corrosion Inspection

Post-tensioning duct grouting was inspected for any enviromental moisture ingress. Effective prestress force after friction loss was calculated as: 1500.0 * 0.92 = 1410.0 kN. Corrosion rate progression was compared against baseline Study GEO-2024-19A and citation [8].

Appendix A: Geotechnical Borehole Log Summary

Bedrock refusal was encountered at a depth of 18.4 meters below Pier P3 foundation footing.

References

[1] AASHTO LRFD Bridge Design Specifications, 9th Edition, American Association of State Highway Officials.

[2] FHWA-NHI-15-047, Seismic Retrofitting Manual for Highway Structures, Federal Highway Administration.